1 A rod is fixed horizontally. A solid bar of mass M is attached to the rod by two vertical strings so that the bar is horizontal, as shown in Figure 1.1.

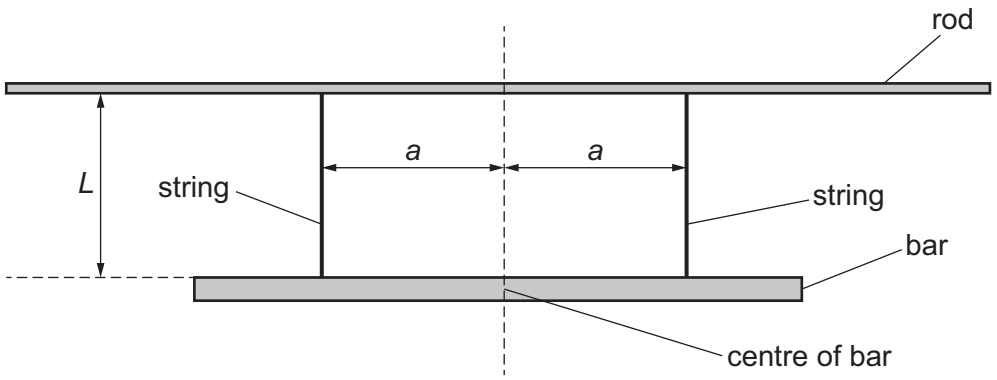

Figure 1.1

The vertical strings are equal horizontal distances a from the centre of the bar.

The separation between the rod and the bar is L.

One end of the bar is displaced horizontally and then released. The bar rotates with oscillations about its centre. The period of the oscillations is T.

It is suggested that T is related to a by the relationship

$$T = \frac{2\pi P}{a^q}\sqrt{\frac{L}{M}}$$

where P and q are constants.

Plan a laboratory experiment to test the relationship between T and a.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine values for P and q.

In your plan, you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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